



JAEBEOM LEE, Ph.D.

Senior Research Scientist

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EDUCATION

Ph.D. in Civil Engineering, August 2020, UNIST, Ulsan, South Korea

- **Dissertation:** Probabilistic Assessment of Bridge Safety by Integrating Measurements with Computational Simulations

B.S. in Civil Engineering, August 2015, UNIST, Ulsan, South Korea

RESEARCH INTERESTS

- [Theoretical Interest] Uncertainty Quantification (UQ)
 - Forward UQ: Sampling-Based & Non-Sampling-Based & Bayesian Methods.
 - Inverse UQ: Bayesian Methods (e.g., Markov Chain Monte Carlo Method).
 - Model UQ: Quantification of Model Bias & Variance.
- [Application #1] Bayesian Deep Learning-Based Nondestructive Testing
- [Application #2] Bayesian Machine Learning-Based Structural Health Monitoring
- [Application #3] Finite Element Reliability Analysis for Lifetime Prediction of Infrastructures

RESEARCH EXPERIENCE

Assistant Professor

March 2025 – Present
Measurement Engineering
University of Science and Technology (UST)

Senior Research Scientist

June 2022 – Present

Nondestructive Metrology Group
Korea Research Institute of Standards and Science (KRISS), South Korea

Visiting Scholar

August 2025 – December 2025
Digital Instrumentation Department (DIN)
CEA-LIST, France (French Alternative Energies and Atomic Energy Commission)
Inviter: Dr. Pierre Calmon

Postdoctoral Researcher

September 2020 – June 2022
Structural Reliability & Disaster Risk Laboratory
Ulsan National Institute of Science and Technology (UNIST), South Korea
Advisor: Prof. Young-Joo Lee

Research Advisor

September 2020 – April 2021
THYROSCOPE INC. (Smart Healthcare Startup), South Korea

Research Advisor

September 2020 – June 2022
사단법인 맑은도시 (Corporation Aggregate), South Korea

Graduate Research Assistant

August 2015 – August 2018
Structural Reliability & Disaster Risk Laboratory
Ulsan National Institute of Science and Technology (UNIST), South Korea
Advisor: Prof. Young-Joo Lee

Visiting Student Researcher

January 2015 – February 2015
R4 Laboratory
University of Illinois Urbana-Champaign, US
Advisor: Prof. Eun Jeong Cha

RESEARCH PROJECTS (PRINCIPAL INVESTIGATOR)

자율지능형 비파괴 검사 로봇 탑재용 불확실성 인식 임베디드 인공지능 개발

(Development of Uncertainty-Aware Embedded AI for Autonomous Intelligent Nondestructive Testing Robots)

- Duration: March 2026 – February 2031
- Grants: KRW 660,800,000 / five years

- Funding code: RS-2026-25476039
- Supported by the National Research Foundation (NRF) grant funded by the Korea Government (MSIT)

디지털트윈 기반 시설물 성능 추정 기술 개발

- Duration: January 2025 – December 2029
- Grants: KRW 4,800,000,000 / five years
- Supported by Korea Research Institute of Standards and Science (KRISS)

딥러닝 및 디지털트윈 기반 인프라구조물의 가상 센서 시스템 구축 기법 개발

- Duration: January 2023 – December 2024, January 2026 – December 2027
- Grants: KRW 100,000,000 / three years
- Supported by Korea Research Institute of Standards and Science (KRISS)

디지털트윈의 심층강화학습 기반 자동 업데이트를 통한 인프라 구조물의 수명 예측 기법 개발

(Development of lifetime prediction method for infrastructures using digital twin automatically updated by deep reinforcement learning)

- Duration: March 2021 – February 2026
- Grants: KRW 563,664,000 / five years
- Funding code: NRF-2021R1C1C2008770
- Supported by the National Research Foundation (NRF) grant funded by the Korea Government (MSIT)

AWARDS AND HONORS

Commendation from the President of KRISS (원장 표창)

@ World Metrology Day Ceremony (May 2026)

“Meritorious Service for the Advancement of Metrology Science and Technology (측정과학기술 발전 유공)”

Excellent Paper Winner

@ 2025 Academic Symposium of Computational Structural Engineering Institute of Korea (November 2025)

Best Paper Winner

@ The 3rd Young Engineers Colloquium in East Asia (January 2024)

Excellent Paper Award

@ Korean Society of Civil Engineering 2021 Convention (October 2021)

Third Prize

@ The 1st International Project Competition for Structural Health Monitoring (IPC-SHM 2020) (January 2021)

Third Prize among 30 International teams

Best Paper Award

@ 2019 International Symposium on Sea-Crossing Bridges (October 2019)

Excellent Paper Award

@ Korean Society of Civil Engineering 2015 Convention (October 2015)

Seismic Simulation Test Center Award

@ The Seismic Structural Design Contest (July 2015)

Excellence Award

@ 2013 Ulsan Invention Contest (November 2013)

Excellence Award

@ App Development Contest for Technical Standards (March 2013)

Scholarship for Educational Expenses

March 2011 – August 2015

Grants: KRW 28,020,200

Excellence Entrance Award

High entrance score scholarship for undergraduate school (March 2011)

Funding for visiting UIUC

TEACHING EXPERIENCE

Graduate Teaching Assistant at UNIST

Probability concepts in engineering

Numerical modelling and analysis

Structural reliability

Finite element analysis

PUBLICATIONS: INTERNATIONAL SCI(E)

Published

1. Lee, J., Lee, S., Lee, Y.-J.*, and **Lee, J.*** (2026). Predicting Seismic Floor Response for Nuclear Power Plant Structures with Time-Series Uncertainty Propagation using Attention-Enhanced Multimodal Deep Learning, *Reliability Engineering & System Safety*, 272, 112582.
2. Jeon, M., Choi, W., Park, J.W., **Lee, J.***, and Mun, D.* (2026). Enhancing Chromium Coating Thickness Estimation with Multi-Head Attention LSTM and Data Augmentation. *Scientific Reports*, 16, 8286.
3. Lee, S., Jeon, C.-H., **Lee, J.***, and Lee, Y.-J.* (2026). Prediction of the Flexural Behavior of Corroded Prestressed Concrete Girders: A Probabilistic Multi-Level Approach, *Structural Safety*, 120, 102685.

4. Lee, J., Lee, S., Lee, Y.-J.* , and **Lee, J.*** (2025). Virtual Sensing of Seismic Floor Responses for Rapid Prioritization of Critical Equipment Inspection in Nuclear Power Plants, *Computer-Aided Civil and Infrastructure Engineering*, 40(26), 4669-4688.
5. Lee, S., **Lee, J.***, Jeon, C.-H., and Lee, Y.-J.* (2025). Probabilistic Prediction of Load-Displacement Curves of Corroded Strands, *Developments in the Built Environment*, 22(2025), 100644.
6. **Lee, J.**, Kim, H., Park, J. W., Ko, D., Jang, H., and Choi, W.* (2025). Adaptive Probabilistic Estimation of Chromium Coating Thickness for Nuclear Fuel Cladding using Non-Parametric Regression with ECT Sensors, *Nuclear Engineering and Technology*, 57(8), 103586.
7. Park, J. W., Ko, D., Jang, H., Kim, H., **Lee, J.**, Choi, W.* (2024). Estimation of Cr Coating Thickness for Accident Tolerant Fuel Cladding using Pancake ECT Sensor, *Nuclear Engineering and Technology*, 57(2), 103216.
8. Lee, S., **Lee, J.**, Yoon, S., and Lee, Y.-J.* (2024). Efficient Finite Element Reliability Analysis Employing Sequentially-Updated Surrogate Model for Fragility Curve Derivation, *Structures*, 64, 107246.
9. Kim, M., **Lee, J.**, Lee, K., Jang, J.H., and Lee, Y.-J.* (2024). Data-driven method of probabilistic response prediction for cable-stayed bridges, *Advances in Civil Engineering*, 2024, 16.
10. Lee, S., **Lee, J.***, Kim, M., and Lee, S. (2024). Deep Learning-Based Anomaly Detection in Acceleration Data of Long-Span Cable-Stayed Bridges, *Smart Structures and Systems*, 33(2), 93-103.
11. **Lee, J.**, Jeon, C.-H., Shim, C.-S., and Lee, Y.-J.* (2023). Bayesian Inference of Pit Corrosion in Prestressing Strands Using Markov Chain Monte Carlo Method, *Probabilistic Engineering Mechanics*, 74(2023), 103512.
12. **Lee, J.**, Jeong, S., Lee, J., Sim, S.-H., Lee, K.-C.* , and Lee Y.-J.* (2022). Sensor Data-Based Probabilistic Condition Monitoring of Railway Bridges using Time-History Deflections Induced by High-Speed Trains, *Structural Health Monitoring*, 14759217211063424.
13. **Lee, J.**, Lee, Y.-J.* , and Shim, C.-S.* (2021). A Multi-Scale Framework for Probabilistic Structural Analysis of PSC Girders Considering Pit Corrosion of Prestressing Wires. *Engineering Structures*, 244, 112745.
14. **Lee, J.**, Lee, Y.-J.* , and Shim, C.-S.* (2020). Probabilistic Prediction of Mechanical Characteristics of Corroded Strands. *Engineering Structures*, 203, 109882.
15. **Lee, J.**, Lee, K.-C.* , Sim, S.-H., Lee, J., and Lee, Y.-J.* (2019). Bayesian Prediction of Pre-Stressed Concrete Bridge Deflection Using Finite Element Analysis. *Sensors*, 19(22), 4956.

16. **Lee, J.**, Lee, Y.-J.* , Sim, S.-J., and Cho, S. (2019). A New Framework for Structural System Fragility and Sensitivity Analysis of Concrete Gravity Dams. *KSCE Journal of Civil Engineering*, 23(8), 3592-3605.
17. **Lee, J.**, Lee, K.-C.* , and Lee, Y.-J.* (2018). Long-Term Deflection Prediction from Computer Vision Measured Data History for High-Speed Railway Bridges. *Sensors*, 18(5), 1488.
18. Kim, H., Sim, S.-H., **Lee, J.**, Lee, Y.-J.* , and Kim, J.-M. (2017). Flood Fragility Analysis for Bridges with Multiple Failure Modes. *Advances in Mechanical Engineering*, 9(3), 1-11.
19. **Lee, J.**, Lee, Y.-J.* , Kim, H., Sim, S.-H., and Kim, J.-M. (2016). A New Methodology Development for Flood Fragility Curve Derivation Considering Structural Deterioration for Bridges. *Smart Structures and Systems*, 17(1), 149-165.

PATENTS & SOFTWARE

1. 최원재, 지봉규, 박정원, 승홍민, 하종문, **이재범** (2023). 디스플레이 패널 고속 검사 방법(High-Speed Inspection Method for Display Panels). 출원번호: 10-2023-0151049, 국제출원(PCT): WO2025/095654 A1.
2. 이영주, Sreehari Ramachandra Prabhu, **이재범** (2019). 재료 파괴 특성 분석 장치 및 방법(Apparatus and Method for Analyzing Failure Characteristics of Material). 출원번호: 10-2017-0140450, 등록번호: 10-1982842.
3. 이영주, 이승준, **이재범** (2019). 페럼-마이다스(FERUM-MIDAS). 등록번호: C-2019-031996.
4. **이재범**, 이영주 (2016). 몬테카를로 아바쿠스(Monte Carlo ABAQUS). 등록번호: C-2016-005407.

CONFERENCE PAPERS (INTERNATIONAL)

1. **Lee, J.**, Lee, S., and Yoon, D.-J. (2025). Structural Condition Monitoring Through Information Transferring with Dimensional Expansion. *13th International Conference on Structural Health Monitoring of Intelligent Infrastructure (SHMII-13)*, September 1-5, Graz, Austria.
2. Lee, J., **Lee, J.**, and Lee, Y.-J. (2025). Design of 1D CNN Kernels for Predicting Seismic Response of Nuclear Power Plants. *28th International Conference on Structural Mechanics in Reactor Technology (SMiRT28)*, August 10-15, Toronto, Canada. (AWARDED)
3. **Lee, J.**, Kim, J., Jeon, M., Park, C., and Lee, S. (2025). Probabilistic Anomaly Detection Beyond Observed Environmental Conditions. *The European Safety and Reliability (ESREL) and Society for Risk Analysis Europe (SRA-E) Conference*, June 15-19, Stavanger, Norway.

4. Lee, S., **Lee, J.**, Jeon, C.-H., and Lee, Y.-J. (2025). Probabilistic Performance Curve Estimation of PSC Beams with Corroded Strands Using a Multiscale Approach. *6th International Conference on Uncertainty Quantification in Computational Science and Engineering (UNCECOP2025)*, June 15-18, Rhodes, Greece.
5. Han S., **Lee, J.**, and Kim, T. (2025). Framework for Assessing Flood-Induced Bridge Performance Considering Climate Change. *14th International Conference on Structural Safety and Reliability (ICOSSAR25)*, June 1-6, Los Angeles, California, USA.
6. **Lee, J.**, Park, Y., and Kim, S. (2024). Enhancing Performance of Probabilistic Structural Condition Monitoring through Transfer Learning. *The 8th Asian-Pacific Symposium on Structural Reliability and Its Applications (APSSRA2024)*, September 29-October 2, Ulsan, Korea.
7. **Lee, J.** (2024). Artificial Intelligence-Based Structural Condition Monitoring. *The 2024 Emerging Scientist Workshop (ESW2024)*, August 26-28, NIM Changping Campus, Beijing, China.
8. **Lee, J.**, Choi, W., Seung, H.M., and Ha, J.M. (2024). Structural Condition Monitoring using Deep Gaussian Process Regression. *The 34th European Safety and Reliability Conference (ESREL2024)*, June 23-27, Cracow, Poland.
9. **Lee, J.**, Choi, W., Seung, H.M., and Ha, J.M. (2024). Intuitive Structural Condition Monitoring through Time-Domain Anomaly Detection. *20th World Conference on Non-Destructive Testing (20th WCNDT)*, May 27-31, Incheon, Korea.
10. Kim, M., **Lee, J.**, and Lee, Y.-J. (2024). Probabilistic Prediction on Multiple Structural Responses for Cable Bridges based on Measurement Data. *The 3rd Young Engineers Colloquium in Ease Aisa (YEC) by The International Association for Bridge and Structural Engineering (IABSE)*, Jan. 19–20, Seoul, Korea.
11. **Lee, J.**, Kim, H., Lee, S., Lee, H., and Kim, J. (2023). Comparative Study on Optimization Methods in Finite Element Model Updating. *The 33rd European Safety and Reliability Conference (ESREL2023)*, Sep. 3–7, Southampton, United Kingdom.
12. Lee, S., **Lee, J.**, Yoon, S., and Lee, Y.-J. (2023). Efficient Finite Element Reliability Analysis Employing Surrogate Model for Seismic Fragility Curve Derivation. *14th International Conference on Application of Statistics and Probability in Civil Engineering (ICASPI4)*, Jul. 9–13, Dublin, Ireland.
13. **Lee, J.**, Jeon, C.-H., Shim, C.-S., and Lee, Y.-J. (2022). Probabilistic Estimation of Pit Corrosion in Prestressing Strands for Prestressed Concrete Bridges. *The 32nd European Conference on Safety and Reliability Conference (ESREL2022)*, Aug. 28–Sep. 1, Dublin, Ireland.
14. **Lee, J.**, and Kim, H. (2022). Probabilistic Finite Element Model Updating by Data Fusion of Acceleration and Angular Velocity. *11th International Conference on Structural Health Monitoring of Intelligent Infrastructure (SHMII-11)*, Aug. 8–12, Montreal, QC, Canada.

15. **Lee, J.**, Shim, C.-S., and Lee, Y.-J. (2022). Probabilistic finite element modeling of PSC beam considering pit corrosion. *The 13th International Conference on Structural Safety and Reliability (ICOSSAR 2021-2022)*, June 20-24, Shanghai, China.
16. **Lee, J.**, Jeon, C.-H., Shim, C.-S., and Lee Y.-J. (2022). Probabilistic Estimation of Pit Corrosion in Prestressing Strands for Prestressed Concrete Bridges, *The 32nd European Safety and Reliability Conference (ESREL2022)*, Aug. 28-Sep. 1, Dublin, Ireland.
17. **Lee, J.**, Jeong, S., Lee, J., Sim, S.-H., Lee, K.-C., and Lee Y.-J. (2021). Time-History Deflection Monitoring of Railway Bridges Using Gaussian Process Regression, *The 2nd ZHITU Symposium on Advances in Civil Engineering*, Sep. 29-29, Ulsan, Korea.
18. **Lee, J.**, Jeong, S., Lee, J., Sim, S.-H., Lee, K.-C., and Lee Y.-J. (2021). Time-History Deflection Monitoring of Railway Bridges Subjected to High-Speed Trains, *Asia Pacific Conference of the Prognostics and Health Management 2021 (PHMAP 2021)*, Sep. 8-11, Jeju, Korea.
19. **Lee, J.**, Shim, C.-S., and Lee, Y.-J. (2020). Probabilistic Inference on Corrosion Configurations Based on Tensile Test Results of Corroded Strand, *The 2020 World Congress on Advances in Civil, Environmental, & Materials Research (ACEM20)/The 2020 Structures Congress (Structures20)*, Aug. 25-29, Seoul, Korea.
20. **Lee, J.**, Lee, K.-C., Sim, S.-H., Lee, J., and Lee, Y.-J. (2019). Vertical Deflection Prediction of PSC Bridge in Construction Stage Using Gaussian Process Regression, *2019 International Symposium on Sea-Crossing*, Oct. 24-25, Seoul, Korea.
21. **Lee, J.** and Lee, Y.-J. (2019). Bayesian Prediction of Deflection Based on Measurement Data for Cable-Stayed Bridges, *2019 International Symposium on Sea-Crossing Bridges*, Oct. 24-25, Jeollanam-do, Korea.
22. **Lee, J.**, Lee, Y.-J., and Shim, C.-S. (2019). Probabilistic Prediction of Ultimate Strength and Strain of Corroded Steel Strands, *29th European Safety and Reliability Conference (ESREL 2019)*, Sep. 22-26, Hannover, Germany.
23. **Lee, J.**, Shim, C.-S., and Lee, Y.-J. (2019). Probabilistic Range Prediction of Ultimate Strength of Corroded Seven-Wire Strands, *The 2019 World Congress on Advances in Structural Engineering and Mechanics (ASEM19)*, Sep. 17-21, Jeju, Korea.
24. **Lee, J.**, Lee, K.-C., Sim, S.-H., and Lee, Y.-J. (2019) Probabilistic Prediction of Vertical Deflection for Bridges using Gaussian Process Regression, *9th International Conference on Structural Health Monitoring of Intelligent Infrastructure (SHMII-9)*, Aug. 4-7, St. Louis, MO, USA.
25. **Lee, J.**, Lee, K.-C., and Lee, Y.-J. (2019). Probabilistic Prediction of Vertical Deflection for High-Speed Railway Bridges using a Gaussian Process, *13th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP13)*, May 26-30, Seoul, Korea.

26. Lee, Y.-J., **Lee, J.**, and Lee, S. (2018). Probabilistic Fatigue Life Prediction Employing an Advanced Crack Propagation Model, *The 9th International Conference on Bridge Maintenance, Safety and Management (IABMAS 2018)*, July 9-13, Melbourne, Australia.
27. **Lee, J.**, Lee, Y.-J., Kim, H., and Sim, S.-H. (2017). Finite Element Reliability Analysis for Deriving Flood Fragility Curve of Deteriorated Bridges Considering Scour, *The 12th International Conference on Structural Safety and Reliability (ICOSSAR12)*, Aug. 6-10, TU Wein, Vienna, Austria.
28. **Lee, J.**, Lee, Y.-K., Kim, H., and Sim, S.-H. (2016). Flood Fragility Analysis for Multiple Failure Modes of Bridges by Finite Element System Reliability Analysis, *2016 World Congress on Advances in Civil, Environmental, and Materials Research (ACEM16)*, Aug. 28-Sep. 1, Jeju, Korea.